ENGR 6311
Vibrations in Machines and Structures
Winter 2026

Course Instructor:
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Office Hours:
Announced on Moodle.

Course Calendar Description:
Vibrations of discrete systems: Single-Degree of Freedom (SDOF) and Multi-Degree of Freedom (MDOF) systems; continuous systems: bars, beams, membranes and plates with various boundary conditions; mode superposition; energy methods; Rayleigh-Ritz Method; condensation techniques; applications to machine components, rotor bearing systems, vehicle and aerospace structures. Project on selected topics is an integral part of the course.

Prerequisites or co-requisites: None.
This course is a pre-requisite to ENGR 7311, Mech 6231, Mech 6301, Mech 6311, Mech 6351.

Specific Knowledge and Skills Needed for this Course:
For this course, a strong working knowledge in kinematics, ordinary differential equations and linear algebra is essential. The contents of undergraduate courses in Dynamics and Mechanics of Materials will be used frequently in lectures. The course begins with a brief review (3-4 weeks) of the main contents of an undergraduate-level course in mechanical vibrations. A significant portion of the assignments is based on code-based simulations.

Course materials:
Primary textbook:
- B.H. Tongue, Principles of Vibrations, Oxford University Press, 2nd edition, 2002.
  https://concordiauniversity.on.worldcat.org/oclc/176049841

Additional resources will be provided through Moodle.

Contents:
The course covers fundamental concepts on vibrations of mechanical systems, including analysis of vibrating systems and design of vibration mitigation strategies. The course builds up on the contents of an introductory undergraduate course on mechanical vibrations and covers theoretical and numerical methods for predicting the vibration response of discrete and continuous mechanical systems under general loading conditions. The focus of the lectures is on analytical methods. Several assignments and projects are provided throughout the course to reinforce the concepts and demonstrate the application of computational and analytical tools to investigate vibration problems in machinery and structures.

Tentative Course Schedule

- Vibrations of systems with one degree of freedom: Chapters 1 to 3
- Vibrations of systems with multiple degrees of freedom: Chapter 4
- Vibrations of continuous systems (strings, bars, beams): Chapters 5 & 6
- Special topics

Grading Scheme

- Course project: 50%
- Homework assignments: 50%

The instructor reserves the right to consider additional bonus marks (up to 10% of the total grade) for classwork, exemplary performance or other exceptional circumstances if deemed appropriate.

Additional projects or assignments for boosting the grade will not be provided. Plan accordingly.

Students need to secure at least 50% of the mark for course project and 50% of the overall marks for assignments to pass the course.

Other Information

Homework assignments

- The majority of the homework assignments are designed in the style of small simulation projects.
- If you want to use any programming language or software other than Matlab, Octave or Python, you need to confirm with the instructor before submitting the homework assignment. Otherwise, you may lose the grade for that assignment. Microsoft Excel is strongly discouraged for programming.
- If you need to access university computers (for example, to use Matlab), follow the information in this link: https://www.concordia.ca/ginacody/aits/support/faq/connect-from-home.html
- In rare occasions that late submissions of homework assignments is acceptable, it would lead to a penalty on the grade. Please plan accordingly.

Examination and Grading

- There will be no make-up for any graded component of the course.
- To request re-grading of any graded component other than the final exam, students need to submit a written form (available on Moodle). A re-grade request is only accepted until 7 days after a grade is announced. Only one re-grade request is accepted per graded component.
- The instructor reserves the right to conduct an individual oral examination for any graded component of the course to verify a student's response to specific questions and/or to re-evaluate a student’s knowledge on any aspect or component of the course. This re-evaluation can override any component of the student’s grade.
- The course letter grade is an indicator, among other factors, of a student’s performance and relative standing in the class. There is no fixed relation between overall marks and letter grades.

Delivery of course material

- There will no course notes available for this course.
- Enquiries through Moodle or Teams chat will not be answered. Please use email instead.

Health and Safety Guidelines

All health and safety rules specific to this course can be found in the lab manual. General health and safety instructions and available health and safety trainings are discussed at:
Safety Programs - Concordia University (https://www.concordia.ca/campus-life/safety/general-safety.html)

Should your course have additional information about health and safety guidelines/training, please insert here.

ON CAMPUS RESOURCES

Student may find the full list of available resources at:

- Important Services & Resources - Concordia University (https://www.concordia.ca/students/services.html)

- A-Z Student Services - Concordia University (https://www.concordia.ca/students/services/a-z.html)

Following section is optional

While they are included in above mentioned two links, following list includes frequently searched services and resources.

1. Academic Resources - Concordia University
   a. Academic integrity - Concordia University
   b. Undergraduate Calendar - Concordia University
   c. Graduate Calendar - Concordia University (For Graduate Classes)
   d. Academic dates – Undergraduate - Concordia University
   e. Important dates – Graduate - Concordia University (For Graduate Classes)
   f. Learning services - Concordia University
   g. Advocacy - Concordia University
2. Health & Wellness - Concordia University
   a. Access Centre for Students with Disabilities - Concordia University
   b. Mental health - Concordia University
3. Safety & security - Concordia University
   a. Security services - Concordia University
   b. Environmental Health and Safety - Concordia University
   c. Safety Training - Concordia University
4. International students - Concordia University